Grade: Check Plus

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1. Mean & SD of the stock price.

x	f(x)	x · f(x)
20	0.05	1.00
21	0.06	1.26
22	0.07	1.54
23	0.08	1.84
24	0.09	2.16
25	0.12	3.00
26	0.15	3.90
27	0.13	3.51
28	0.10	2.80
29	0.10	2.90
30	0.05	1.50

mean is = $25.41 ✓ Q1

SD = subtract mean from each price, square the difference, multiply by probability, add all together

$20-25.41=(-5.41)^2=29.27(0.05)=1.46$
$21-25.41=(-4.41)^2=19.45(0.06)=1.17$
$22-25.41=(-3.41)^2=11.63(0.07)=0.81$
$23-25.41=(-2.41)^2=5.81(0.08)=0.46$
$24-25.41=(-1.41)^2=1.99(0.09)=0.18$
$25-25.41=(-0.41)^2=0.17(0.12)=0.02$
$26-25.41=(0.59)^2=0.35(0.15)=0.05$
$27-25.41=(1.59)^2=2.53(0.13)=0.33$
$28-25.41=(2.59)^2=6.71(0.10)=0.67$
$29-25.41=(3.59)^2=12.89(0.10)=1.29$
$30-25.41=(4.59)^2=21.07(0.05)=1.05$

7.50

$SD=\sqrt{7.50}=\$2.74$

2. a The probability stock is w/in 1 SD of the mean of $25.41 w/a SD of $2.74.

$22.67 \le x \le 28.15$, can only be 23, 24, 25, 26, 27, 28
$(.08)+(.09)+(.12)+(.15)+(.13)+(.10)=\boxed{0.67}$ ✓ Q2a

2b. $2\times\$2.74 +/- \$25.41 = \boxed{\$19.93 \le x \le \$30.89}$ 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30
$(.05)+(.06)+(.07)+(.08)+(.09)+(.12)+(.15)+(.13)+(.1)+(.1)+(.05)=\boxed{1.0}$ ✓ Q2b

3. 20, 21, 22, 23, 24, 25
.05, .06, .07, .08, .09, .12 = $\boxed{.47}$ ✓ Q3

4. 26, 27, 28, 29, 30
.12, .15, .13, .10, .10 = $\boxed{.53}$ ✓ Q4

5. $0(.47)+1(.15)+2(.13)+3(.10)+4(.10)+5(.05)=\boxed{1.36 \text{ mean}}$

v	P(V=v)		$(v-1.36)^2$	Product
0	0.47	×	1.85	0.87
1	0.15	×	0.13	0.019
2	0.13	×	0.41	0.05
3	0.10	×	2.69	0.26
4	0.10	×	6.97	0.697
5	0.05	×	13.24	0.66

2.57

$\sqrt{2.57}=\boxed{\$1.60}$ standard deviation ✓ Q5